

# Reliable Robotic Manipulation Through Evidence-Grounded Control

Across identical simulation protocols, optimized PID produced the strongest aggregate reliability while no controller dominated every condition.

### Problem and research question

Reliable manipulation requires more than nominal trajectory tracking: controllers must remain interpretable under payload, configuration, disturbance, noise, and actuator changes.

### Method

A 2-link arm compares manual PID, Mamdani Fuzzy-PID, and bounded optimization-assisted PID under identical 0.001 s simulation, trajectory, torque-limit, threshold, and paired-seed protocols.

### Key results

Evidence snapshot: 7.283% nominal objective reduction; 10/13 deterministic scenarios passed; 30/30 vs 0/30 isolated-noise vs combined-stress trials

|                                              |                                                |                                                                  |
|----------------------------------------------|------------------------------------------------|------------------------------------------------------------------|
| <b>7.283%</b><br>nominal objective reduction | <b>10/13</b><br>deterministic scenarios passed | <b>30/30 vs 0/30</b><br>isolated-noise vs combined-stress trials |
|----------------------------------------------|------------------------------------------------|------------------------------------------------------------------|

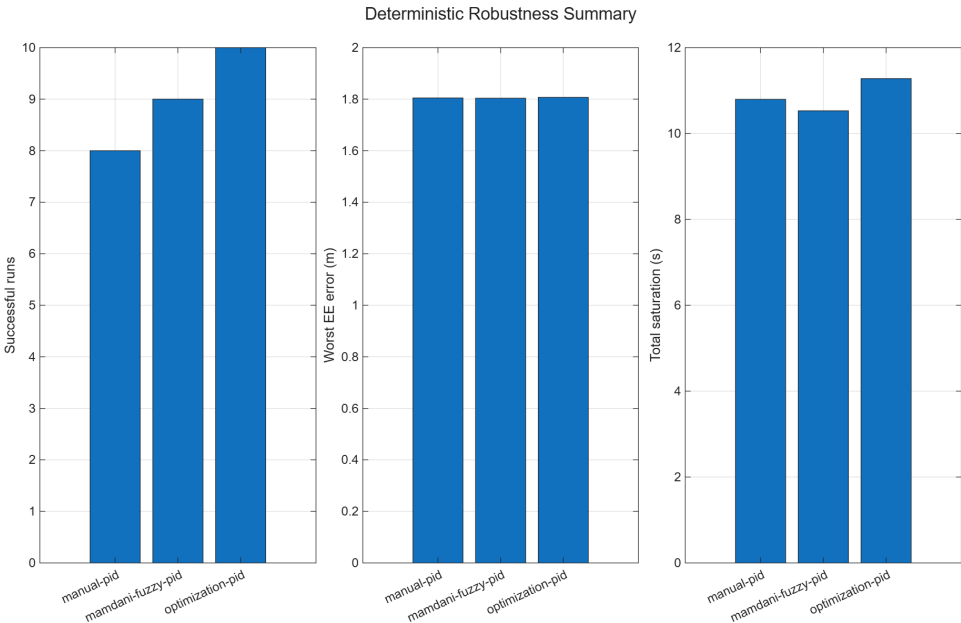


Figure. Deterministic stress testing separates aggregate controller reliability while preserving failed cases.

### Why this matters

The project supplies a transparent low-level execution layer that can sit beneath future perception, language, and task-planning systems.

### Simulation scope

Simulation only; no hardware fidelity, collision avoidance, perception, online safety supervisor, or end-to-end embodied-AI implementation is claimed.

### Limitations and next steps

Validate on hardware, add calibrated sensing uncertainty, and introduce runtime safety monitoring before deployment claims.

Sources: docs/report/technical\_report.md | results/report/report\_evidence.json | docs/report/PAPER\_CLAIM\_AUDIT.md